



Common Tools

WIDGET API VERSION 1.4.0

Overview

Common tools are JavaScript and CSS utilities that ship with iCUE that can be used in your widgets to enhance the user experience with widget configuration.

HOW TO USE COMMON TOOLS

To use any common tool in your widget, you must **copy the `common` folder** (or the specific tool subfolder you need) from iCUE widgets folder (`<<iCUE install dir>>/widgets/`) into your widget's root directory. Your widget references these files via relative paths (e.g., `common/tools/media_viewer/MediaViewer.js`), so the files must be present alongside your `index.html`.

Your widget folder should look like this:

```
MyWidget/
├─ index.html
├─ manifest.json
└─ common/
    └─ tools/
        └─ media_viewer/
            ├─ MediaViewer.js
            └─ MediaViewer.css
```

MediaViewer

Displays and transforms images and videos selected via a `media-selector` property. Handles zoom, pan, and rotation exactly as configured in the iCUE media editor.

File: `tools/media_viewer/MediaViewer.js` + `tools/media_viewer/MediaViewer.css`

When to use: Any widget with a `backgroundImage` (or similar) `media-selector` property.

Constructor

```
const viewer = new MediaViewer(options);
```

Option	Type	Required	Description
<code>container</code>	<code>HTMLElement</code>	Yes	Element that receives the media
<code>onMediaLoaded</code>	<code>function</code>	No	Callback when media loads successfully
<code>onMediaError</code>	<code>function</code>	No	Callback on load failure
<code>debug</code>	<code>boolean</code>	No	Enable verbose logging

Methods

Method	Description
<code>loadMedia(params)</code>	Load and display media with transforms
<code>clear()</code>	Remove current media, hide container
<code>destroy()</code>	Clean up all resources

`loadMedia(params)` **Parameters**

Parameter	Type	Description
<code>path</code>	string	Path/URL to the media file
<code>baseWidth</code>	number	Original media width (from <code>media-selector</code> output)
<code>baseHeight</code>	number	Original media height (from <code>media-selector</code> output)
<code>scale</code>	number	Scale factor (default: <code>1</code>)
<code>positionX</code>	number	X offset in pixels
<code>positionY</code>	number	Y offset in pixels
<code>angle</code>	number	Rotation in degrees

Note: Parameters are `baseWidth` / `baseHeight` — iCUE's `media-selector` output uses these exact field names.

Container CSS Requirements

```
#media-background {  
  width: 100%;  
  height: 100%;  
  position: relative; /* Required – MediaPlayer uses absolute children */  
  /* Do NOT add overflow: hidden – breaks MediaPlayer's translate positioning */  
  /* Do NOT add visibility: hidden – MediaPlayer manages its own visibility */  
}
```

Supported Formats

Type	Extensions
Images	<code>jpg</code> , <code>jpeg</code> , <code>png</code> , <code>gif</code> , <code>bmp</code> , <code>webp</code> , <code>svg</code> , <code>ico</code>
Videos	<code>mp4</code> , <code>webm</code> , <code>ogg</code> , <code>mov</code> , <code>avi</code> , <code>mkv</code>

Example

```
<!-- Inline in <head> -->
<style>
    .position-container {
        position: absolute;
        transform-origin: top left;
    }
    .scale-container {
        transform-origin: top left;
    }
    .rotation-container {
        transform-origin: center center;
    }
    .rotation-container img,
    .rotation-container video {
        display: block;
        object-fit: contain;
    }
</style>
<script>
    class MediaViewer {
        /* ... inline full source ... */
    }
</script>
```

```
const viewer = new MediaViewer({ container: document.getElementById("media-
background") });

function applyBackground() {
    if (typeof backgroundImage === "undefined" || !backgroundImage) {
        viewer.clear();
        return;
    }
    viewer.loadMedia({
        path: backgroundImage.pathToAsset,
        baseWidth: backgroundImage.baseWidth,
        baseHeight: backgroundImage.baseHeight,
        scale: backgroundImage.scale,
        positionX: backgroundImage.positionX,
        positionY: backgroundImage.positionY,
        angle: backgroundImage.angle,
    });
}
```

See the `media-selector` control documentation for full integration guidance.

💡 TIP

Remember to copy the `common/tools/media_viewer/` folder from your iCUE installation into your widget directory before packaging.